Lab Report – EE-170L Computer Programming (Lab 12)

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LAB	12 cpp
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Objectives

- To understand computer memory and variables.
- To learn pointer declaration in C++.
- To apply referencing (&) and dereferencing (*) operators.
- To explore pointer arithmetic with arrays.

Theory

1. Computer Memory

· Memory is divided into **bytes**, each with a unique **address**.

· A variable reserves a block of memory depending on its type (e.g., `int` = 4 bytes on 32-bit systems).

· The compiler maps variable names to memory addresses.

2. Pointers

· A pointer is a variable that stores the address of another variable.

· Declaration syntax:

```
dataType *ptr;
```

· Referencing operator (&) → gives the address of a variable.

· Dereferencing operator (*) → accesses or modifies the value stored at the address.

3. Pointer Arithmetic

· Pointers can be incremented or decremented to move across memory addresses.

- Useful for traversing arrays without using indices

Lab Tasks

Task 1: Pointer Declaration and Initialization

```
#include <iostream>     //tells the compiler to include input output stream
library

using namespace std;    // tells the compiler to incude standard namespace


int main() {                // main function begins from here

    int num = 25;           // declare integer variable

    int *ptr;               // declare pointer to int

    ptr = &num;             // store address of num in ptr

    cout << "Value of num: " << num << endl;       // display num directly

    cout << "Value via ptr: " << *ptr << endl;      // display value using pointer

    cout << "Address of num: " << ptr << endl;     // display address stored in ptr

    return 0;                                      // end program

}
```

Task 2: Pointer Arithmetic with Arrays

```
#include <iostream>      //tells the compiler to include input output stream
library

using namespace std;     // tells the compiler to incude standard namespace


int main() {                                          // main function begins from here

   int numbers[5] = {10, 20, 30, 40, 50};       // array declaration and
initialization

   int *ptr;                                                // declare pointer

   ptr = numbers;                                            // point to first element of
array



   cout << "Array elements using pointer:" << endl;    // output prompt

   for(int i = 0; i < 5; i++) {                                // for loop that runs 5 times

      cout << *(ptr + i) << " ";                // access elements using pointer
arithmetic

   }

   cout << endl;                                            // moves compiler to next line

   return 0;                                                 // end program

}
```

Results

- Learned how pointers store addresses of variables.
- Demonstrated dereferencing to access values through pointers.
- Applied pointer arithmetic to traverse arrays without indices.
- Observed how memory addresses change as the pointer moves across array elements.